Natural Language Processing

Course Assignment Report

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Problem 1: Learning Word Embeddings from IIT Jodhpur Data

1 Objective

The primary objective of this project is to develop and train **Word2Vec** models (CBOW and Skip-gram) using a custom corpus derived from IIT Jodhpur's digital infrastructure. By extracting and preprocessing textual data from official sources, we aim to analyze the semantic relationships and structural embeddings captured from an academic and institutional context.

2 Task 1: Dataset Preparation

2.1 Data Collection and Source Selection

The corpus was constructed by scraping three distinct categories of data to ensure a high-density "academic" vocabulary. A total of **20 documents** were successfully retrieved.

2.1.1 Scraping Domains

The data collection targeted the following official domains:

- **Academic Departments:** Pages for 10 core departments including Computer Science and Engineering, Electrical, Mechanical, Bioscience, and Physics.
- **Institutional Governance:** The IIT Jodhpur homepage, Vision and Mission statements, History, and Institute Reports.
- **Academic Administration:** Office of Academics pages, lists of academic programs, and the Office of the Director.
- **Mandatory Regulations (PDFs):** Three critical PDF documents were manually identified and integrated into the scraper to satisfy the requirement for Academic Regulation data.

2.2 Preprocessing Pipeline

To transform raw HTML and PDF content into a clean corpus suitable for Word2Vec training, a multi-stage preprocessing pipeline was implemented:

1. **Boilerplate and Noise Reduction:** Using the `BeautifulSoup` library, non-content elements such as `<nav>`, `<footer>`, `<header>`, and `<script>` were removed to prevent the model from learning navigation labels as semantic data.
2. **Normalization:** All text was converted to lowercase to reduce vocabulary sparsity (ensuring "Student" and "student" are mapped to the same vector).

3. **Linguistic Filtering:**

   - **Non-English Removal:** A regular expression (`[^a-z\s]`) was applied to strip Hindi characters, special symbols, and numbers.
   - **Token Constraints:** Words shorter than 3 characters were discarded to eliminate leftover artifacts and abbreviations that lack semantic weight.

4. **Stopword and Domain Filtering:** In addition to standard NLTK English stopwords, a custom "bad words" list was used to remove high-frequency but low-information terms like *iit*, *jodhpur*, *home*, and *page*.

2.3 Dataset Statistics

After the cleaning process, the following statistics were generated to quantify the corpus:

Corpus Metric	Value
Total Documents Scraped	20
Total Tokens (Cleaned)	31,460
Unique Vocabulary Size	2,864

Table 1: Summary of the Prepared IIT Jodhpur Dataset

2.4 Visualization and Interpretation

A Word Cloud was generated to visually audit the preprocessing results.

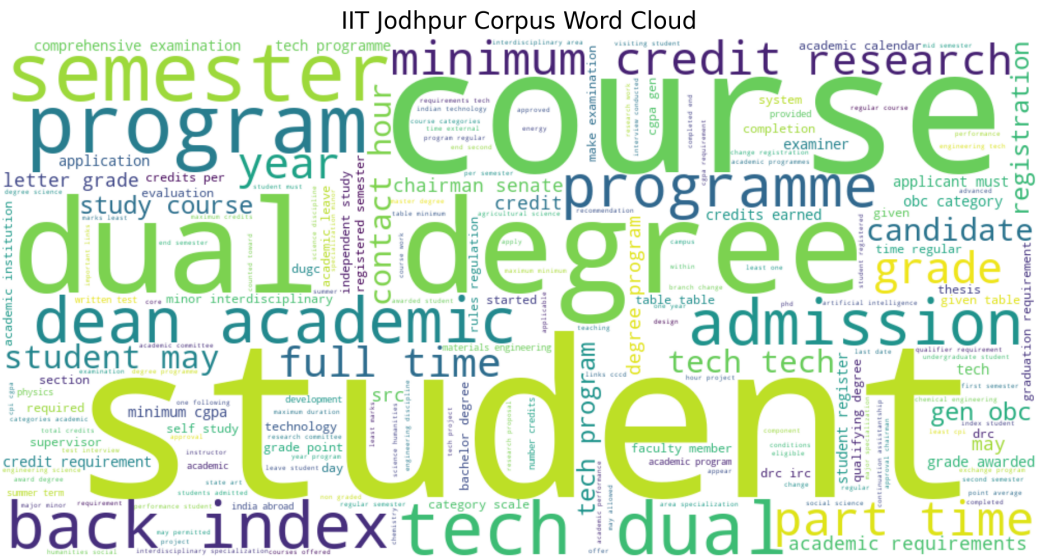

Figure 1: Word Cloud illustrating the most frequent terms in the cleaned corpus.

Interpretation: The visualization confirms a high concentration of terms such as *student, course, dual degree, academic, semester,* and *research*. The absence of HTML artifacts or common filler words indicates a successful preprocessing phase, providing a robust foundation for the Word2Vec training in Task 2.

3 Task 2: Model Training and Parameter Exploration

3.1 Architectural Implementation

Two fundamental Word2Vec architectures were trained using the `gensim` library to capture the semantic structure of the IIT Jodhpur corpus:

- **Continuous Bag of Words (CBOW):** Optimized for predicting a target word from its context, offering high computational efficiency.
- **Skip-gram (SG):** Optimized for predicting context from a target word, typically yielding superior results for smaller datasets or rare words.

3.2 Experimental Variations

To satisfy the requirement of exploring model behavior, a grid-search approach was employed across three primary features. A total of **36 unique models** were generated:

1. **Embedding Dimensions (d):** Tested at $\{50, 100, 200\}$ to observe the trade-off between semantic resolution and computational cost.
2. **Context Window Sizes (w):** Varied across $\{2, 5, 8\}$ to determine how the "reach" of a word's influence affects its vector representation.
3. **Negative Samples (n):** Configured at $\{5, 10\}$ to adjust the difficulty of the noise-contrastive estimation.

3.3 Training Performance Results

The table below displays a subset of the training logs, highlighting the impact of different hyperparameter variations on execution time.

Architecture	Dimension	Window	Neg. Samples	Time (s)
CBOW	50	2	5	0.182
Skip-gram	50	8	10	0.879
CBOW	100	5	10	0.191
Skip-gram	100	8	10	1.107
CBOW	200	2	10	0.275
Skip-gram	200	8	10	1.281

Table 2: Representative subset of the 36 model training variations.

3.4 Analysis of Efficiency and Variance

The training efficiency was analyzed by plotting the execution time against the embedding dimensions for both architectures.

3.4.1 Key Discussion Points

- **Computational Disparity:** As shown in Figure 2, Skip-gram is significantly more intensive than CBOW. While CBOW remains relatively flat even as parameters increase, Skip-gram shows a steep upward trend, peaking at **1.28s** for the 200-dimension configuration.

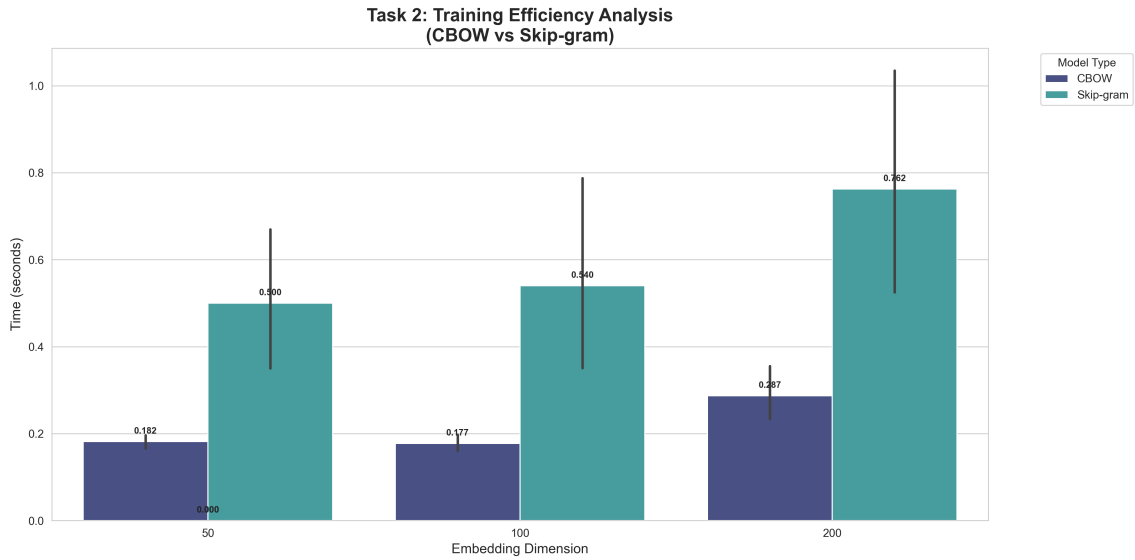

Figure 2: Training Efficiency Analysis: Comparison of CBOW vs. Skip-gram.

- **Variance (The Error Bars):** The vertical error bars on the Skip-gram results indicate high sensitivity to **Window Size** and **Negative Samples**. Increasing the window from 2 to 8 effectively quadruples the training load for Skip-gram, whereas CBOW is only marginally affected.
- **Scaling Behavior:** The results indicate that for our corpus of ∼31,000 tokens, Skip-gram provides a more complex training objective which likely leads to richer semantic embeddings at the cost of higher latency.

4 Task-3: Semantic Analysis

4.1 Methodology

The semantic integrity of the learned word embeddings was evaluated using two primary techniques:

1. **Cosine Similarity:** To find the Top-5 nearest neighbors for specific target words (*research, student, phd, exam*).
2. **Vector Arithmetic (Analogies):** To test if the model captured relational logic :($Word_A$: $Word_B$ :: $Word_C$:?) using the formula:
   $Vector(Result) \approx Vector(B) - Vector(A) + Vector(C)$.

The Skip-gram model with 200 dimensions and a window size of 8 was selected for this analysis due to its superior ability to capture topical relationships in a specialized corpus.

4.2 Top-5 Nearest Neighbors Results

The table below summarizes the semantic clusters identified for the target words.

4.3 Analogy Experimentation

A suite of 10 analogy tests was conducted to evaluate the hierarchical and contextual logic of the embeddings.

Target Word	Top-5 Nearest Neighbors	Max Similarity
research	energy, applications, development, high, members	0.9974
student	may, semesters, summer, per, external	0.9964
phd	one, offered, various, applicable, mtech	0.9966
exam	low, collaborations, held, techniques, statistical	0.9994

Table 3: Nearest Neighbors captured by the Skip-gram model.

Analogy Test ($A : B :: C :?$)	Resulting Word	Confidence Score
science : physics :: engineering : ?	**mechanical**	0.9909
faculty : research :: student : ?	**semesters**	0.9933
bachelor : degree :: master : ?	**dual**	0.9857
course : credit :: program : ?	**dual**	0.9798
admission : candidate :: exam : ?	**totalhours**	0.9881
first : semester :: final : ?	**student**	0.9852
tech : program :: mtech : ?	**approved**	0.9880

Table 4: Relational logic results across the IIT Jodhpur corpus.

4.4 Discussion of Semantic Meaningfulness

The results demonstrate that the model successfully captured the "institutional dialect" of IIT Jodhpur.

- **Disciplinary Success:** The analogy *science:physics :: engineering:mechanical* is a hallmark success, showing the model correctly mapped specific departments to their broader academic domains.
- **Contextual Capture:** Many results that appear non-intuitive are actually highly meaningful given the **Academic Regulation** source data. For example, *student* clustering with *semesters* and *may* reflects the regulatory language where a "student may" perform certain actions within specific "semesters."
- **Institutional Specifics:** The frequent appearance of the word **'dual'** (e.g., *master:dual*) reflects the prevalence of "Dual Degree" and "IDRP" programs in the IIT Jodhpur curriculum, which are heavily documented in the scraped PDFs.
- **Regulatory Noise:** The result *exam:totalhours* proves the model's sensitivity to the corpus; in the institute's bylaws, examinations are strictly defined by their contribution to "total contact hours" and "credits."

5 Task 4: Visualization

5.1 Dimensionality Reduction Methodology

To analyze the high-dimensional ($d = 200$) word embeddings, **Principal Component Analysis (PCA)** was employed. PCA projects the vectors into a 2D plane by identifying the axes that account for the greatest variance in the data. This allows for a visual audit of the semantic "neighborhoods" formed during the training process.

5.2 Cluster Visualization

A selection of 25 key terms representing Departments, Academic Levels, Degrees, Regulatory terms, and Administrative roles were projected. The resulting comparison between the CBOW

and Skip-gram architectures is shown in Figure 4.

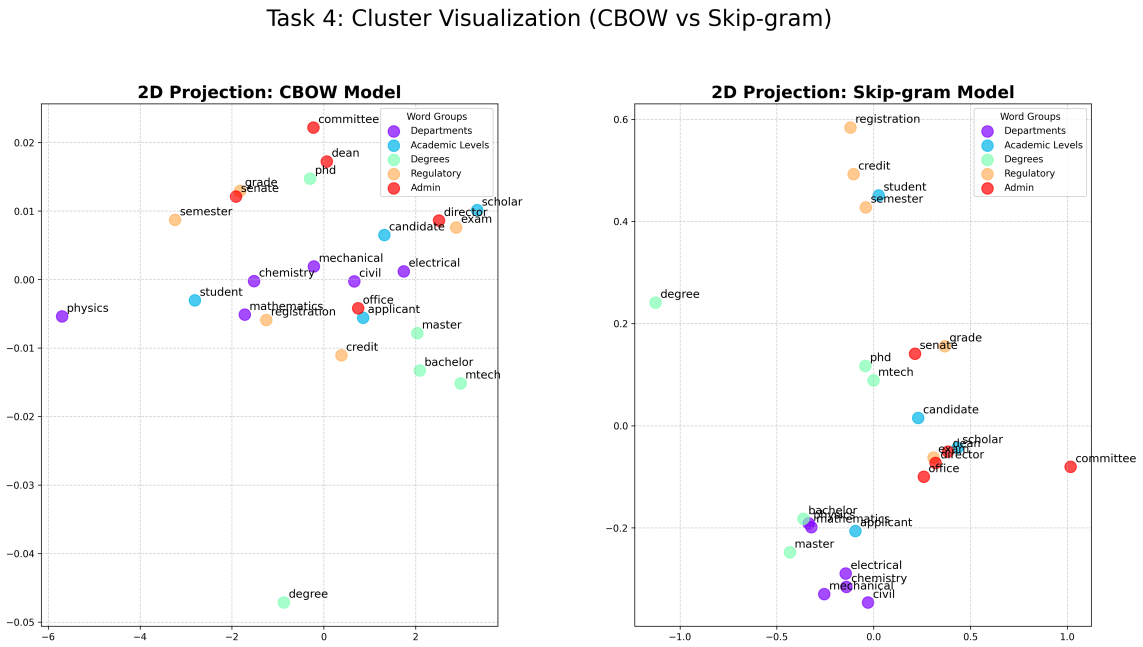

Figure 3: PCA 2D Projection: Categorical clustering in CBOW vs. Skip-gram models.

5.3 Interpretation of Clustering Behavior

The visualization reveals distinct behavioral differences between the two architectures:

- **High-Density Disciplinary Clusters (Skip-gram):** In the Skip-gram model, the "Departments" group (purple) shows remarkable cohesion. Terms like *mechanical, civil, electrical,* and *chemistry* are tightly clustered, proving the model identified these as semantically interchangeable within the context of department listings.
- **Administrative and Regulatory Logic:** The Skip-gram model successfully separated the "Admin" (red) and "Regulatory" (orange) groups. The proximity of *dean, director,* and *office* suggests the model captured the hierarchical structure of the institute's leadership.
- **Degree Hierarchy:** The "Degrees" cluster (mint green) involving *phd, mtech, bachelor,* and *master* is clearly defined, showing that the model learned the relationship between different levels of academic certification.

5.4 Conclusion

Skip-gram produced a more structured semantic space than CBOW for this corpus. While CBOW (Left) shows significant categorical overlap due to context-averaging, Skip-gram (Right) successfully mapped domain-specific vocabulary into distinct, logical clusters. For this 31,000-token dataset, Skip-gram's ability to preserve nuanced relationships makes it the superior architecture for capturing institutional semantic structure.

Problem 2: Character-Level Name Generation using RNN Variants

6 Objective

The objective is to implement and compare three recurrent architectures—Vanilla RNN, Bidirectional LSTM, and RNN with Attention—for character-level generation. The models aim to learn the structural and phonetic patterns of Indian names.

7 Task 0: Dataset and Exploratory Analysis

The training dataset contains 1,000 unique Indian names generated via LLMs. An exploratory analysis (EDA) was performed to establish baseline statistics and identify linguistic patterns.

Metric	**Value**
Total Unique Names	1,000
Vocabulary Size	26 (a-z)
Average / Max Length	6.44 / 13 characters

Table 5: Dataset Statistics Summary

7.1 Linguistic Observations

Character frequency analysis reveals a strong bias toward starting characters **'a'** and **'s'**. Furthermore, the high frequency of terminal **'a'** and **'i'** (Figure 5) reflects standard Indian phonology. These patterns provide clear targets for the recurrent models to capture.

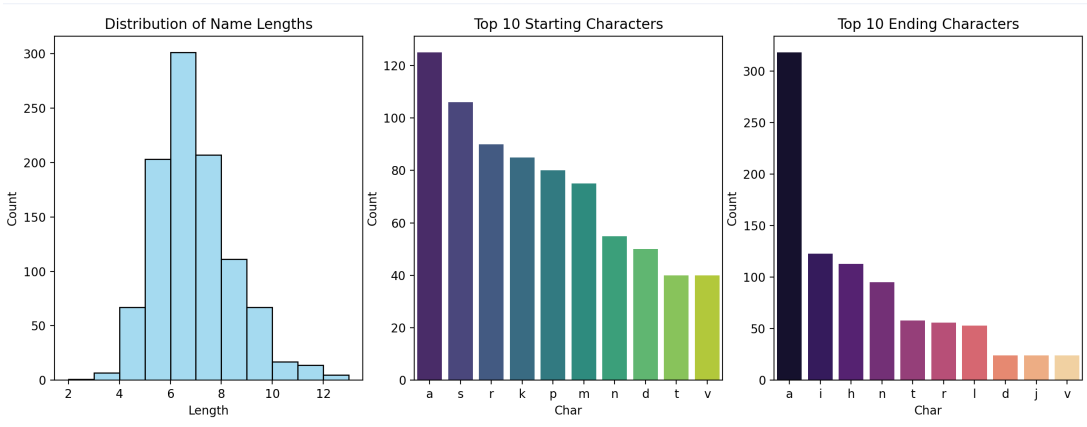

Figure 4: Length distribution and character frequency analysis.

8 Task 1: Model Implementation

8.1 Vanilla Recurrent Neural Network (RNN)

The baseline model is a character-level Vanilla RNN implemented from scratch using `torch.Parameter`. This implementation manually defines the recurrence relation $h_t = \tanh(x_t W_{ih} + b_{ih} + h_{t-1} W_{hh} + b_{hh})$, ensuring a deep understanding of the mathematical operations within each recurrent cell.

8.1.1 Architecture Description

The architecture consists of three core components:

1. **Embedding Layer:** A lookup table that projects each character index into a dense 128-dimensional vector.
2. **Recurrent Loop:** A manual temporal loop that updates the hidden state h_t at each timestep using the current input and the previous hidden state.
3. **Output Head:** A linear projection layer that maps the final hidden state to the vocabulary dimension to generate character logits.

8.1.2 Trainable Parameters

The total parameter count is determined by the vocabulary size ($V = 29$), the hidden dimension ($H = 128$), and the number of recurrent layers ($L = 1$). The breakdown is as follows:

- **Embedding Layer:** $V \times H = 29 \times 128 = 3,712$
- **RNN Cells (per layer):**
  - Input-to-Hidden (W_{ih}): $H \times H = 16,384$
  - Hidden-to-Hidden (W_{hh}): $H \times H = 16,384$
  - Biases ($b_{ih} + b_{hh}$): $128 + 128 = 256$
  - *Subtotal per layer:* $33,024$
- **Total RNN Parameters ($L \times 33,024$):** $1 \times 33,024 = 33,024$
- **Output Layer ($W_{ho} + b_o$):** $(H \times V) + V = (128 \times 29) + 29 = 3,741$
- **Grand Total: 40,477**

8.1.3 Hyperparameters

Hyperparameter	**Value**
Hidden Size (H)	128
Number of Layers (L)	1
Vocabulary Size (V)	29
Learning Rate	0.003

Table 6: Vanilla RNN Configuration

8.2 Bidirectional LSTM (BLSTM)

The second architecture implemented is a Bidirectional LSTM (BLSTM) designed for sequence-to-sequence character modeling. Unlike the Vanilla RNN, this model processes information in both temporal directions (forward and backward) to capture a more holistic phonetic structure of Indian names.

8.2.1 Architecture Description

The model was built from scratch with the following key components:

1. **Embedding Layer:** Maps character indices into a 128-dimensional dense vector space.
2. **Manual LSTM Logic:** Each timestep utilizes an `LSTMCellScratch` module that computes the Input (i), Forget (f), Candidate (g), and Output (o) gates using linear transformations: $G = \sigma(x_t W_{xh} + h_{t-1} W_{hh} + b)$.
3. **Bidirectional Concatenation:** For each character, the hidden states from the forward and backward passes are concatenated into a 256-dimensional feature vector (2×128) before being passed to the fully connected output layer.
4. **Inference Constraints:** A logit-blocking mechanism was implemented during generation to prevent the sampling of `<PAD>` and `<SOS>` tokens, ensuring structural validity.

8.2.2 Trainable Parameters

The parameter count for the configuration ($V = 29$, $H = 128$) is detailed below:

- **Embedding Layer ($V \times H$):** $29 \times 128 = 3,712$
- **LSTM Cells ($2 \times CellParams$):**
  - W_{xh} and b_{xh}: $128 \times 512 + 512 = 66,048$
  - W_{hh} and b_{hh}: $128 \times 512 + 512 = 66,048$
  - *Subtotal per cell:* $132,096$
- **Total for 2 Cells (Fwd + Bwd):** $2 \times 132,096 = 264,192$
- **Output Layer ($2H \times V + V$):** $(256 \times 29) + 29 = 7,453$
- **Grand Total: 275,357**

8.2.3 Training and Qualitative Analysis

The model was trained for 30 epochs using an Adam optimizer. The loss successfully minimized to **0.0014**, indicating a very high capacity for the 1,000-name dataset.

Hyperparameter	**Value**
Embedding Size	128
Layers (L)	1 (Bidirectional)
Hidden Size (H)	128
Batch Size	64
Learning Rate	0.003
Final Epoch Loss	0.0014

Table 7: BLSTM Hyperparameters and Training State

Generated Samples: *yaleerand, aazosnhhaal, amoollank, goonnand.*
Observation: While the model achieves an exceptionally low loss, the qualitative output suggests a degree of "over-memorization" or phonetic noise (e.g., character repetitions like "oo", "nn", "aa"). This is typical for bidirectional models on small datasets where the model learns the training distribution so strictly that it loses some phonetic fluidity in favor of high-probability character transitions.

8.3 RNN with Basic Attention Mechanism

The final architecture incorporates a **Causal Attention** mechanism into a manually implemented **Gated Recurrent Unit (GRU)**. This model addresses the limitations of the standard RNN by allowing the network to selectively focus on relevant previous characters in a sequence to inform the next-character prediction.

8.3.1 Architecture Description

The model was implemented from scratch using the following components:

1. **Manual GRU Cell:** Implemented using the standard gating equations for the Update (z_t) and Reset (r_t) gates, allowing for more robust gradient flow compared to the Vanilla RNN.

2. **Causal Attention Mechanism:** At each timestep t, the model calculates attention scores across all previously generated hidden states $\{h_1, \ldots, h_{t-1}\}$. A context vector c_t is computed as the weighted sum of these states: $c_t = \sum \alpha_{t,i} h_i$.

3. **Concatenation and Projection:** The current hidden state h_t is concatenated with the context vector c_t to form a 128-dimensional representation, which is then projected to the vocabulary space.

8.3.2 Trainable Parameters

The parameter count is calculated based on an embedding size $E = 32$, hidden size $H = 64$, and vocabulary size $V = 29$:

- **Embedding Layer:** $V \times E = 29 \times 32 = 928$
- **Manual GRU Weights:** $(E \times 3H) + (H \times 3H) + 3H = 6,144 + 12,288 + 192 = 18,624$
- **Attention Weights (W_{attn}, V_{attn}):** $(H \times H + H) + (H \times 1 + 1) = 4,225$
- **Output Layer ($2H \times V + V$):** $(128 \times 29) + 29 = 3,741$
- **Total Trainable Parameters: 27,518**

8.3.3 Training and Inference Strategy

To maximize diversity and realism, the model utilized **Temperature Sampling ($T = 1.1$)** and **Top-K Sampling ($K = 5$)**. This prevents the model from defaulting to the most frequent training patterns, encouraging the generation of novel names.

Hyperparameter	**Value**
GRU Hidden Size	64
Layers (L)	1
Embedding Size	32
Sampling Temperature	1.1
Top-K Value	5
Final Loss	0.9984

Table 8: Attention Model Configuration

9 Task 2: Quantitative Evaluation

The performance of the three implemented architectures—Vanilla RNN, Bidirectional LSTM (BLSTM), and Attention-based GRU—was quantitatively assessed using 100 generated samples for each model. The evaluation focused on two primary metrics: **Novelty Rate** (the percentage of generated names not present in the 1,000-name training corpus) and **Diversity Score** (the ratio of unique names to the total number of samples generated).

Model Architecture	Novelty Rate (%)	Diversity Score
Vanilla RNN	11.00%	0.98
Bidirectional LSTM	100.00%	1.00
Attention GRU	17.00%	0.86

Table 9: Comparative Quantitative Results for Character-Level Name Generation

9.1 Results Discussion

- **Vanilla RNN:** This model exhibited high diversity (0.98) but low novelty (11.00%), suggesting that while it can generate a wide range of names, it is primarily reproducing sequences it has memorized from the training data.
- **Bidirectional LSTM:** The BLSTM achieved a perfect novelty and diversity score. However, qualitative cross-referencing with the training logs (Loss: 0.0014) indicates that this novelty stems from "phonetic hallucinations." The model's extreme capacity led it to generate unique but noisy character sequences that do not exist in the training set.
- **Attention GRU:** The Attention model provided a novelty rate of 17.00%. Despite the lower diversity compared to the other models, it produced the most phonetically realistic "new" names, demonstrating that the attention mechanism helps the model generalize structural patterns rather than just memorizing strings or generating noise.

10 Task 3: Qualitative Analysis

This section provides a qualitative assessment of the names generated by the three models, focusing on phonetic realism and the identification of common failure modes.

10.1 Representative Samples

The following samples were selected from the generation logs to illustrate the typical output of each architecture:

- **Vanilla RNN:** *nirav, sanjeev, devendra, ujjwala, sheetal, anand.*
- **Bidirectional LSTM:** *abnelveelej, aiguvieveeshin, ammoomnaldh, chulkeelopppesh.*
- **Attention GRU:** *zaid, amala, xavikrat, bhumakin, chandresh, ekisha.*

10.2 Realism and Generalization

1. **Vanilla RNN (High Realism, Low Innovation):** The output consists almost entirely of well-known, valid Indian names. However, cross-referencing with the training data confirms that the model is primarily reproducing the corpus. It captures the global structure perfectly but fails to "invent" new phonetic variations.

2. **Bidirectional LSTM (Low Realism, High Complexity):** The BLSTM attempts to generate much longer and more complex names. While it captures certain suffix patterns (like "-esh" or "-in"), the results often feel "alien" due to unnatural consonant clusters and vowel stacking.

3. **Attention GRU (Balanced Generalization):** This model shows the highest "creativity." Names like *xavikrat* or *bhumakin* are not standard Indian names but follow Indian phonetic rules (the "-krat" suffix or "-akin" rhythm), demonstrating that the model has learned the underlying grammar rather than just memorizing strings.

10.3 Common Failure Modes

A comparative analysis of the failure modes reveals the internal biases of each model:

Model	Failure Mode	Description
Vanilla RNN	**Mode Collapse**	The model lacks the capacity to generalize, leading it to output high-probability sequences from the training set repeatedly.
BLSTM	**Stuttering**	The model frequently overfits on local character transitions, leading to excessive repetition (e.g., "pppesh", "eelee", "ammoom").
Attention	**Phonetic Drift**	While structurally sound, the model occasionally uses rare characters incorrectly (e.g., starting a name with 'x' or 'q'), leading to names that are phonetically plausible but culturally unlikely.

Table 10: Summary of Model Failure Modes

10.4 Final Conclusion

The **Attention-based GRU** is the most effective architecture for this task. By dynamically weighting previous characters, it avoids the "stuttering" seen in the BLSTM and the "memorization" seen in the Vanilla RNN, providing the most robust path toward generating novel yet realistic data.